

QueryBook / LEL — Description of the Drawings

Companion to the Drawings (FIG. 1 – FIG. 30). Provides the Brief Description of the Drawings and the Reference Numerals keys. Draft — for a provisional patent application; formal drawing compliance (37 CFR 1.84) and claim language to be finalized by counsel.

Brief Description of the Drawings

FIG. 1 is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

FIG. 2 is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

FIG. 3 is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

FIG. 4 is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

FIG. 5 is a block diagram of the seven-stage Speech Output Pipeline.

FIG. 6 is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

FIG. 7 is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two, per Section VII.

FIG. 8 is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

FIG. 9 is a diagram of the queryBook system architecture (Prime Directive as Domain 0).

FIG. 10 is a diagram of the fact Unit — atom, envelope, and content-derived identity.

FIG. 11 is a diagram of the fact-certification authority hierarchy.

FIG. 12 is a diagram of the uFCS substrate — three-store architecture.

FIG. 13 is a diagram of the authored tour and exhibition sequencing.

FIG. 14 is a diagram of the fQL — joint structure and trust execution.

FIG. 15 is a diagram of the verifying machine-generated text against certified records.

FIG. 16 is a diagram of the federation — cross-boundary re-validation.

FIG. 17 is a diagram of the fact-level erasure by external-key destruction.

FIG. 18 is a diagram of the universal security architecture.

FIG. 19 is a diagram of the universal honeypot saturation and data firehose.

FIG. 20 is a diagram of the deployment topologies over an invariant common layer.

FIG. 21 is a diagram of the adaptive response engine — directive fixed before generation.

FIG. 22 is a diagram of the user-owned agent system with governance and kill switch.

FIG. 23 is a diagram of the deterministic multi-language regeneration.

FIG. 24 is a diagram of the semantic compression, expansion, colorization and upscaling.

FIG. 25 is a diagram of the queryBook-enhanced TCP/IP hybrid protocol packet.

FIG. 26 is a diagram of the neuromorphic cognitive layer — attractor convergence.

FIG. 27 is a diagram of the fact Unit — confidence and evidence model.

FIG. 28 is a diagram of the uFCS packet structure — serialized Fact Unit record.

FIG. 29 is a diagram of the uFCS packet framing — content-addressed block storage.

FIG. 30 is a diagram of the fQL — query anatomy and worked example.

Reference Numerals

Language Expression Layer (FIG. 1 – FIG. 8)

No.	Element	No.	Element
100	QueryBook LEL system	451	negative-transfer detection & retraining
110	Sub-Language Priming Layer (SLPL)	500	Speech Output Pipeline
120	Language Expression Layer (LEL)	510	lexical selection
130	Transition Gate	520	grapheme-to-phoneme conversion
140	Phase Two semantic-communicative learning	530	coarticulation modeling
150	Multilingual Delta Acquisition module	540	prosody generation
160	Speech Output Pipeline	550	register & affect calibration
165	natural speech output	560	self-monitoring feedback loop
170	co-grounding interface	570	neural vocoder rendering
180	Federated Query Language (FQL)	580	natural speech waveform
190	Universal Federated Connectivity System (UFCS)	600	co-grounded integration
195	federated data sources	610	natural-language input
200	SLPL	620	LEL analysis
210	acoustic phonetic processing	630	FQL operations
211	signal-level feature embedding	640	UFCS execution
220	phonics / phonemic awareness	650	federated sources
221	phoneme-boundary segmenter	660	results
230	communication-science foundations	670	render to NL / speech in input language
231	communicative-salience weight	680	language-agnostic resolution
240	prosody & rhythm processing	700	Language Lab prototype embodiment
241	prosodic contour vector	710	corpus source
250	information-theoretic grounding	720	Phase-One structural agent
251	information weighting	730	structural analyzer
260	neurolinguistic chunking	731	grapheme inventory
261	chunk segments	732	rule-seeded G2P
270	licensed multimedia input	733	syllabifier
300	developmental learning sequence	734	lexical-stability metric
310	Phase One (structural-acoustic)	735	co-occurrence density metric
311	sound discrimination	736	prosody proxy
312	syllabic segmentation	740	Fact Unit store
313	word-as-acoustic-object	750	Transition-Gate metrics
314	semantics suppressed	751	one-way latch
320	Transition Gate (one-way)	760	phase agents on shared manager
321	phonemic-inventory completeness $\geq \tau_1$	761	Phase-1 agent (implemented)
322	lexical stability $\geq \tau_2$	762	Phase-2 agent — deterministic dictionary + store grounding (implemented)
323	co-occurrence density $\geq \tau_3$	763	Phase-3 agent (declines — spec-only)
324	pattern saturation $< \epsilon$	764	Phase-4 agent (declines — spec-only)
330	Phase Two (semantic-communicative)	800	deterministic Phase-Two semantic grounding
331	word-to-referent grounding	810	Phase-One vocabulary (learned words)
332	communicative discovery	820	deterministic grounder (per word)

No.	Element	No.	Element
333	emergent grammar / syntax	830	bundled public-domain dictionary source
400	Delta Acquisition Model	831	dictionary meaning Fact Unit (means, trust 0.9)
410	reused L1 foundation	840	self-grounding store-lookup source
420	L2 delta detector	841	store-link Fact Unit (grounded_by_fact)
430	targeted training of novel elements	850	Fact Unit store (provenance, domain language)
440	multilingual semantic alignment	860	excluded large-language-model meaning source
450	acquisition acceleration flywheel	870	optional future dictionary-gated suggestor

QueryBook / UFCS Platform (FIG. 9 – FIG. 30)

The reference numerals labeled in the platform figures, in the 100–2200 series. Coordinate and dimension values in the source figures are not reference numerals and are excluded. Counsel/draftsperson to confirm consistency on conversion to a non-provisional.

No.	Element	No.	Element
108	re-validate	1104	fake entry points
112	PRIME DIRECTIVE — SAFETY SUBSTRATE · DOMAIN 0	1106	infinite-apparent-volume response
200	FACT ENVELOPE	1210	INVARIANT COMMON LAYER — PRIME DIRECTIVE (D0) · TRANSFER MATRIX · PATH
202	NUCLEUS	1212	AGENT RUNTIME — user-owned agents execute UNDER governance in every to
204	→ semantic_fingerprint	1400	definition + register
206	→ FUID	1402	commissioner-bounded authority
208	$\alpha / \beta \cdot \text{score} \cdot \text{credible interval}$	1404	template $\cap$ entitlement
210	ref · chain_index · ed25519 sig	1406	ACTION
212	authority_class · trust_score	1408	execute
214	anchor · ingested · attestation	1412	commissioner halts the instance at any time
216	classification · ACL	1500	ingested once
218	typed relations to other Fact Units	1502	language-independent single knowledge source
300	self-certifying	1504	TARGET LANGUAGES
302	score → {t}	1600	pixels / waveform
408	SUBSTRATE RELATIONSHIP TYPES	1602	extract meaning
510	author-selected order	1604	compact Fact Units
512	READER PLAYBACK CONTROL	1800	energy $E = -\frac{1}{2} \sum w_{ij} a_i a_j$
600	structure pattern + trust $\geq \tau$	1802	stable equilibrium
700	extracted claim	1804	capacity ceiling $\alpha_c \approx 0.138$
704	diff object?	1806	noisy cue
708	cite higher trust	1900	α supporting β contradicting
800	DOMAIN A	1902	Beta(α, β) over truth
802	DOMAIN B	1904	point estimate
804	fingerprint · signature · authority	1906	[lo, hi] at 95%
806	re-check sig + trust before admission	1908	independent corroboration
900	encrypted payload	1912	CONFIDENCE TREND
902	held off-system	2020	EXEMPLAR PACKET
904	key destroyed	2100	PACKET STREAM
906	payload now inert	2102	dedupe by fingerprint
1000	Twelve defensive layers	2104	~1 GB gzip/bz2
1002	Five postures	2106	content-addressed lookup
1004	PREDICTION	2210	RESULT
1102	saturates all ports		